

Hitesh Rasineni

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ABOUT ME

CS undergraduate working at the intersection of Machine Learning, statistics, and High Energy Physics, with experience in density estimation and clustering for particle-physics data. Focused on advancing density estimation techniques and applying them to Beyond the Standard Model (BSM) physics.

EDUCATION

VIT-AP University, Amaravati, Andhra Pradesh, India **September 2023 – September 2027**
B.Tech, CSE (AI & ML) CGPA: 8.05 / 10

Sree Vidyanikethan International School, Tirupati, India **2023**
CBSE (10+2) Grade: 91.6 / 100

SELECTED PROJECTS

The Unlikely Lab — Personal Research Publication Platform the-unlikely-lab.github.io/the-unlikely-lab
Code: github.com/The-Unlikely-Lab/the-unlikely-lab

- Designed and deployed a full research-publication platform that hosts complete, typeset HTML versions of arXiv-style physics papers (sections, numbered equations, figures, tables, bibliography) alongside an MDX-driven content system for research notes and project pages.
- Built an automated LaTeX-to-HTML paper build pipeline with Pandoc, including MathML/KaTeX rendering, figure/table validation reports, and per-paper static generation; the site is a statically exported Next.js (React, TypeScript, Tailwind CSS) application deployed via GitHub Actions CI/CD to GitHub Pages.

Cloud-Native ML Training Platform *FastAPI, Docker, Next.js, PyTorch*
Code: github.com/HiteshRasineni/Cloud-Native-ML-Training-Platform-for-Scientific-Data

- Built a cloud-native ML experiment-orchestration platform for HEP data with a working end-to-end vertical slice: a FastAPI backend (PostgreSQL/Alembic + Redis) validates specs, resolves dataset references against a registry, and enqueues jobs; the scheduler runs a QUEUED → SCHEDULING → RUNNING → COMPLETED/FAILED lifecycle (with retries) via a pluggable executor (Docker now, Kubernetes stubbed), launching workload containers that report per-epoch metrics to MLflow + Redis; the Next.js/TypeScript frontend submits experiments and renders loss curves through a backend MLflow proxy.
- Separated storage responsibilities so PostgreSQL owns lifecycle/scheduling state, the dataset registry and artifact index, MLflow (SQLite) owns params, metrics and model artifacts, and MinIO owns datasets, checkpoints and logs; a pluggable worker workload registry runs a from-scratch RealNVP-style normalizing flow in PyTorch (architecture fully spec-driven) alongside a no-op workload, all orchestrated by a docker-compose stack with pytest suites in the backend and worker.

RESEARCH EXPERIENCE

Dark Matter Mono-Z (Density Estimation)

Leptonic Mono-Z Dark Matter Search with Neural Spline Flows

H. Rasineni, B. Chebrolu. arXiv preprint, 2026 (submitted for journal publication). *Leptonic decay channel*. DOI: 10.48550/arXiv.2607.13771

Code: github.com/HiteshRasineni/Leptonic-Mono-z-CMS2015-DarkMatter-Search

- Trained five neural spline flows on 2.32 fb^{-1} of CMS Run 2015D data ($\mu\mu/ee$ channels, 37-dim features) and set 95% CL signal-strength limits via an SR+VR profile-likelihood fit: $\mu < 0.018$ (scalar), 0.036 (vector), 0.050 (axial-vector).

Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching

H. Rasineni, B. Chebrolu. arXiv preprint, 2026. *Hadronic decay channel*. DOI: 10.48550/arXiv.2609.02923

Code: github.com/HiteshRasineni/CMS2015DarkMatterSearch-HTMHT-

- Built the MadGraph5_aMC@NLO → Pythia 8 → Delphes pipeline (1.44M selected events) with a condi-

tional flow-matching background model; projected expected significances of 2.89σ , 7.62σ , and 7.41σ , with extra-jet kinematics carrying 53–71% of the discriminating power.

Likelihood-Based Unsupervised Anomaly Detection in CMS Dijet Events

B. Chebrolu, H. Rasineni, P.A. Immadi (equal contribution). arXiv preprint, 2026.

DOI: 10.48550/arXiv.2609.06686

Paper: the-unlikely-lab.github.io/the-unlikely-lab/research/dijet-anomaly

- Unsupervised, model-independent anomaly search on 378,278 CMS Run 2011B HT events ($\sqrt{s} = 7$ TeV): a neural spline flow learned the SM multijet background from jet-substructure and event-shape observables; the 99.9th-percentile low-likelihood tail was validated with mass-decorrelation, permutation-null, and stability tests (overlap $> 88\%$ across seeds, rank correlation > 0.95).

Improving Discovery-Significance Stability in Higgs Event Classification

J.J. Pujari, P.A. Immadi, H. Rasineni, T. Bikku, R.S. Puppala. *Discover Artificial Intelligence* (Springer Nature), 2026.

DOI: 10.1007/s44163-026-01683-5

- Supervised contrastive pre-training with a parallel FT-Transformer + XGBoost ensemble reduced fold-to-fold variance in Approximate Median Significance ($AMS = 3.74$) versus focal-loss training, enabling more stable threshold selection.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript / TypeScript (React, Next.js)

ML/Data: PyTorch, scikit-learn, OpenCV, Pandas, Matplotlib, Density Estimation (Normalizing Flows, Likelihood Ratios)

Web/Platform: Next.js, React, TypeScript, Tailwind CSS, MDX

Tools/Infra: Git, GitHub Actions (CI/CD), Docker, Pandoc, L^AT_EX, uproot, Google Colab

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals** (Credential)
- **AWS Academy Graduate – Cloud Foundations (Training Badge)** (Credential)
- **WorldQuant University, Applied AI Lab – Deep Learning for Computer Vision** (YOLOv8, MTCNN, Inception-ResNetV1) (Credential)
- **IBM Machine Learning Specialist (Professional V1)** (EDA, ensembles, transfer learning) (Credential)
- **NVIDIA DLI – Fundamentals of Deep Learning** (Credential)
- **NVIDIA DLI – Transformer-based NLP** (Credential)